VAISHNAVI SANTOSHI BATCHU

Senior Data Scientist | Generative AI & Agentic AI Engineer | LLMs · RAG · Multi-Agent Systems · AI Backend Engineering

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PROFESSIONAL SUMMARY

Senior Data Scientist and Generative AI / Agentic AI Engineer with 4+ years designing, deploying, and scaling production-grade AI systems across Generative AI, document intelligence, computer vision, and AI-first backend services. Deep expertise building Large Language Model (LLM) applications, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent orchestration (LangGraph, Google ADK, MCP, tool calling) on Google Cloud Platform (GCP) and AWS, integrated through production APIs, FastAPI/Flask backends, and containerized (Docker) deployments. Experienced working client-facing and cross-functionally to translate enterprise requirements into deployed AI solutions — reducing manual effort by up to 85%, lifting processing accuracy to 97-99%, and driving measurable operational efficiency at 99%+ production availability. 5x GCP- and AWS-certified, with a strong ability to translate complex AI/ML and Agentic AI capabilities into clear, actionable outcomes for enterprise stakeholders.

CORE COMPETENCIES

Data Science • Machine Learning • Artificial Intelligence (AI) • Generative AI (GenAI) • Large Language Models (LLMs) • Retrieval-Augmented Generation (RAG) • Agentic AI & Multi-Agent Systems • Model Context Protocol (MCP) • Tool Calling / Function Calling • Prompt Engineering • Natural Language Processing (NLP) • Computer Vision • Deep Learning • Vector Search & Vector Databases • AI Backend & API Development • MLOps / LLMOps • Model Deployment • CI/CD • Cloud Computing (GCP & AWS) • Client-Facing Solution Delivery • Data Analysis • Predictive Modeling • Stakeholder Management

TECHNICAL SKILLS

Languages: Python (Advanced), SQL (Advanced)

Generative AI & LLMs: Claude, Gemini, PaLM 2, GPT, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-Tuning, Chain-of-Thought Prompting, Few-Shot Prompting

Agentic AI & Orchestration: LangGraph, Google ADK, LangChain, Model Context Protocol (MCP), Tool Calling / Function Calling, Multi-Agent Systems, Autonomous Task Execution, Adaptive Routing, Function Orchestration

Natural Language Processing (NLP): spaCy, NLTK, BERT, GPT, Hugging Face Transformers, Semantic Search

Machine Learning: CNN, Transformers, Regression, Logistic Regression, K-Means, KNN, Decision Tree, Random Forest, Feature Engineering, Model Training & Evaluation

Computer Vision: YOLO, OpenCV, Image Segmentation, Image Deduplication

AI Backend, APIs & Deployment: FastAPI, Flask, Docker, Containerization, REST API Integration, Microservices, CI/CD Pipelines, Model Deployment, Observability & Monitoring

Google Cloud (GCP): Vertex AI, Vertex AI Studio, Document AI, Vertex AI Vector Search, Cloud Run, Cloud Functions, BigQuery, Cloud Storage, Google ADK, Gemini Enterprise, Discovery Engine

AWS: Bedrock, Textract, Lambda, S3, Amazon Rekognition, Serverless Architecture

Databases & Vector Search: PostgreSQL, MySQL, BigQuery, Vector Databases, Semantic / Vector Search, Cosine Similarity

MLOps / LLMOps: CI/CD, Model Monitoring & Logging, Containerization, Vector Database Pipelines, Production Deployment

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn

WORK EXPERIENCE

Google (via Relanto) Pune, Maharashtra
Senior Data Scientist *Jan 2025 – Present*

- Led architecture and delivery of enterprise-scale Generative AI / agentic AI automation platforms, combining Google Document AI, Gemini LLMs, and Retrieval-Augmented Generation (RAG) for context-aware document understanding and precise knowledge retrieval at scale.
- Reduced task-handling overhead by 40%+ in complex enterprise processes by designing multi-agent orchestration workflows in LangGraph and Google ADK, enabling adaptive routing, autonomous task execution, and dynamic decision-making.
- Cut information-lookup time by 50% by building client-facing AI copilots and conversational assistants — integrating Vertex AI Vector Search, Gemini Enterprise, Google Docs API, PostgreSQL, and MySQL for semantic search, AI-powered form completion, and document querying.
- Integrated LLMs with 5+ enterprise APIs and structured data systems using LangChain, Model Context Protocol (MCP), tool calling, and function orchestration, scaling production-grade GenAI backend services.
- Sustained 99%+ production availability by deploying cloud-native AI applications on GCP Cloud Run with scalable backend architectures, CI/CD pipelines, and observability tooling.

- Cut manual reporting effort by 70% by engineering automated dashboard and reporting agents that query PostgreSQL databases directly, improving operational throughput.
- Lifted response accuracy and factual grounding by 20% across enterprise AI applications by refining prompt engineering practices, vector database pipelines, and semantic retrieval.

Searce Inc Pune, Maharashtra
*Machine Learning Engineer**Aug 2023 – Dec 2024*

- **Financial Document Analysis:** Built an end-to-end document processing pipeline on AWS Bedrock (Claude 3 Sonnet) with custom adapters for varied formats; achieved 99.9% system uptime via a serverless AWS Lambda architecture and cut manual processing time by 85%.
- **Intelligent Document Processing:** Led development of a Document AI solution on GCP, delivering a 95% reduction in processing time and a 60% improvement in process efficiency through robust extraction pipelines with advanced error handling.
- **KYC Processing:** Designed a hybrid ML architecture combining YOLO and spaCy with a proprietary coordinate-mapping system for field-level text extraction, reaching 97% accuracy on identity-document processing.
- **Retail Analytics:** Built a Gemini-powered product analysis system using OpenCV for image preprocessing; drove a 25% gain in inventory efficiency and contributed to 15% sales growth through AI-driven insights.
- **AI-Driven Looker Dashboards:** Built a RAG system using Vertex AI LLMs and vector search for autonomous LookML code generation; fine-tuned an LLM to 90% accuracy and resolved field-name mismatches via targeted prompt engineering, deployed on GCP with Cloud Storage and Discovery Engine.
- **Recommendation Engine:** Built a multimodal recommendation system using cosine-similarity vector search on Vertex AI and the PaLM 2 API for entity extraction, achieving a 20% uplift in user engagement.
- **Address Parser:** Enhanced an intelligent address-parsing system combining YOLO and the PaLM 2 API for accurate component extraction, delivering 98% field-level accuracy across varied formats.

Searce Inc Pune, Maharashtra
*Analyst ML Engineer**Jul 2022 – Feb 2023*

- **Image Deduplication:** Built an embedding-based image deduplication system that removed 30% of redundant assets from the content library, cutting downstream storage and processing costs by 25%.
- **Document Information Extraction:** Expanded custom document information-extraction models on GCP, improving extraction accuracy by 15% while maintaining production-grade scalability and reliability.

EDUCATION

B.Tech, Data Science & Artificial Intelligence2019 – 2023
ICFAI, Hyderabad

CERTIFICATIONS

- Google Cloud – Professional Machine Learning Engineer
- Google Cloud – Professional Data Engineer
- Google Cloud – Associate Cloud Engineer
- Google Cloud – Generative AI Leader
- AWS Certified Machine Learning – Specialty

COMMUNITY SERVICE

- Volunteer Teaching: Taught foundational concepts to students from underserved backgrounds.
- Plantation Drive: Participated in community tree-plantation and environmental sustainability initiatives.
- Charity & Donations: Organized and collected donations to support underserved communities in Hyderabad.

LANGUAGES

English, Telugu, Hindi